

## 8.05 Surfaces and Partial Differentiation

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| 8.05a<br>8.05b | 3-D surfaces | <p>a) Be able to work with functions of two variables, given either explicitly in the form <math>z = f(x, y)</math> or implicitly in the form <math>g(x, y, z) = c</math>, and understand and use the fact that this equation, and its partial derivatives, relate to a 3-D surface.</p> <p><i>An informal understanding only of how the partial derivatives relate to the surface is required.</i></p> <p><i>Functions <math>f(x, y)</math> will involve sums and products of powers of <math>x</math> and <math>y</math> only. Issues relating to domains and ranges will not be considered beyond the appreciation that, for example the surface <math>z = \frac{x}{y}</math> has no point at which <math>y = 0</math>.</i></p> | <p>b) Extend their knowledge of surfaces to those defined by functions of more than two variables, and incorporate trigonometric functions, logarithms and exponentials.</p> |
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| OCR Ref.       | Subject Content         | Stage 1 learners should...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Stage 2 learners should additionally...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| 8.05c          | Sections and contours   | <p>c) Be able to sketch sections and contours, and know how these are related to the surface.</p> <p><i>i.e. Sections of the form <math>z = f(a, y)</math> or <math>z = f(x, b)</math> and contours of the form <math>c = f(x, y)</math>.</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 8.05d          | Partial differentiation | <p>d) Be able to find first and second derivatives, including mixed derivatives.</p> <p><i>Learners will be expected to recognise and use both notations for first- and second-order partial derivatives, including mixed ones.</i></p> <p><i>e.g. <math>\frac{\partial f}{\partial x}</math>, <math>\frac{\partial^2 f}{\partial x \partial y}</math> and <math>f_x</math>, <math>f_{xy}</math>.</i></p> <p><i>Includes the Mixed derivative theorem; namely, that <math>\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}</math> or <math>f_{xy} = f_{yx}</math> for suitably well-defined, continuous functions <math>f</math>.</i></p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 8.05e<br>8.05f | Stationary points       | <p>e) Understand and be able to apply the concept that stationary points of <math>z</math> arise when <math>\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y} = 0</math> (or <math>f_x = f_y = 0</math>) and that these can be maxima, minima or saddle-points.</p> <p><i>Learners should know and understand the basic properties of these stationary points.</i></p> <p><i>Learners will only be required to find stationary points, but will not be required to determine their natures.</i></p>                                                                                                                                                                  | <p>f) Be able to determine, for 3-D surfaces given in the form <math>z = f(x, y)</math>, the nature of maxima, minima and saddle-points by means of the sign of the determinant of the matrix of second partial derivatives (the Hessian Matrix),</p> $\mathbf{H} = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix};$ <p>namely, that:</p> <ol style="list-style-type: none"> <li>1. if <math> \mathbf{H}  &gt; 0</math> and <math>f_{xx} &gt; 0</math>, there is a (local) minimum;</li> <li>2. if <math> \mathbf{H}  &gt; 0</math> and <math>f_{xx} &lt; 0</math>, there is a (local) maximum;</li> <li>3. if <math> \mathbf{H}  &lt; 0</math> there is a saddle-point;</li> </ol> |

| OCR Ref. | Subject Content | Stage 1 learners should... | Stage 2 learners should additionally...                                                                                                                                                                                                           |
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| 8.05g    | Tangent planes  |                            | g) Be able to determine, for 3-D surfaces given in the form $z = f(x, y)$ , the equation of a tangent plane to the surface at a given point $(x, y, z) = (a, b, f(a, b))$ using the formula $z = f(a, b) + (x - a)f'_x(a, b) + (y - b)f'_y(a, b)$ |